

YUE FAN

Ph.D. Candidate in Systems and Control Engineering Operations Management | Decision Sciences | Optimization

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EDUCATION

Ph.D. Candidate in Systems and Control Engineering

Expected May 2027

Case Western Reserve University, Cleveland, OH

Dissertation: Geometric Dispersion Theory for Decision Applications

Advisor: Prof. Behnam Malakooti

B.S. in Electrical & Computer Engineering; Minor in Computer Science

Aug 2016 - Mar 2021

Oregon State University, Corvallis, OR

RESEARCH INTERESTS

Primary: Operations management; decision sciences; operations research and optimization; multi-criteria decision making; decision making under risk; business analytics.

Secondary: Information systems and decision support; interpretable machine learning; AI-enabled and data-driven managerial decision making.

Methods: Constrained optimization, utility and value modeling, rigorous out-of-sample evaluation, statistical inference, and reproducible computational research.

PEER-REVIEWED CONFERENCE PROCEEDING

1. **Fan, Y.**[†], & Wang, J.[†] (2022). *Solar Cells Fabrication Procedures*. In Proceedings of the 3rd International Conference on Electronic Science and Automation Control (ESAC 2022), Advances in Computer, Signals and Systems, 20, 24-31. doi:10.23977/esac2022.004

WORKING PAPERS AND MANUSCRIPTS IN PREPARATION

1. **Fan, Y.**, Wang, H., & Malakooti, B.* *Geometric Dispersion Theory for Single-Stock Decisions: Dominance-Consistent Reservation Prices and Point-in-Time Evidence*. Manuscript in preparation for submission to Decision Analysis.
2. **Fan, Y.**, Wang, H., & Malakooti, B.* *Neural Networks with Geometric Dispersion Theory for Multi-Criteria Decision Making*. Manuscript in preparation.
3. **Fan, Y.**, Wang, H., & Malakooti, B.* *Modeling Wine Preferences by Geometric Dispersion Theory*. Manuscript in preparation.
4. Malakooti, B., Komaki, M., Wang, H., & **Fan, Y.** *A Comparative Analysis of Multi-Criteria and Geometric Dispersion Theory Methods Using Out-of-Sample Validation with Application to Wind Farm Location Selection*. Collaborative manuscript in development; Y. Fan contributed the neural-network comparison experiments.

* Corresponding author. [†] Equal contribution in the ESAC 2022 paper.

RESEARCH PROJECT IN PROGRESS

1. Wang, H., & **Fan, Y.** *Geometric Dispersion Theory as an Interpretable Alternative to SHAP*. Separate collaborative project in progress (working title).

ACCEPTED CONFERENCE PRESENTATION

1. **Fan, Y.**, Wang, H., & Malakooti, B. *Neural Networks with Geometric Dispersion Theory for Multi-Criteria Decision Making*. 2026 INFORMS Annual Meeting, San Francisco, CA, November 1-4, 2026. Abstract accepted; presentation forthcoming.

RESEARCH EXPERIENCE

Graduate Research Assistant - Case Western Reserve University

Sep 2023 - Present

Advisor: Prof. Behnam Malakooti

- Develop dissertation research at the intersection of operations management, decision analysis, optimization, and interpretable predictive modeling; study how mathematically structured decision models can produce stable rating, value, and decision-support outputs beyond ranking-oriented MCDM methods.

Single-Stock Decision Analytics with Geometric Dispersion Theory

2025 - Present

- Developed a single-stock decision framework that separates downside exposure from upside opportunity, derives a two-leg reservation-price rule, and establishes first-order stochastic-dominance consistency for the complete decision rule.
- Built a leakage-controlled point-in-time evaluation using 59 U.S. large-cap stocks, three indices, and 840 weekly observations; the downside component predicts future drawdowns, while the upside component orders future run-ups under horizon-consistent inference.

Neural Networks with Geometric Dispersion Theory for MCDM

2025 - Present

- Developed constrained neural-network/GDT models with simplex-normalized weights; validated them out of sample on wind-farm, product-rating, and health-insurance applications. Lead author of the manuscript and accepted 2026 INFORMS presentation.

Interpretable MCDM Validation for Wine-Quality Prediction

2024 - Present

- Designed a comparative study of constrained GDT variants, classical MCDM methods, and machine-learning baselines under systematic reductions in criteria and training-sample size.
- Implemented 30-repeat shared splits, group-locked holdouts, nested training-size prefixes, train-only transformations, sensitivity checks, and artifact-level reproducibility controls; manuscript development and scientific audit are ongoing.

Neural-Network Benchmarking for Wind-Farm Location Selection

2025 - Present

- Contributed neural-network benchmark design and comparison experiments to a collaborative manuscript on out-of-sample MCDM/GDT validation; contribution is limited to the neural-network extension.

Decision Making Under Risk

Sep 2023 - Apr 2024

- Analyzed choice behavior in the Gonzalez and Wu (1999) dataset using correlation analysis, hypothesis testing, and multilinear, LASSO, and ridge regression; interpreted results through expected utility, cumulative prospect theory, and GDT.

Research Team Lead - Semiconductor Device Physics and Fabrication Seminar

Oct 2021 - Dec 2021

Supervisor: Dr. Munib Wober

- Led a project team studying cleaning and doping procedures and designed a P+/N-type solar-cell structure that contributed to the ESAC 2022 publication.

TEACHING INTERESTS

Operations Management; Operations Research and Optimization; Decision Analysis; Supply Chain and Logistics; Business Analytics; Information Systems and Decision Support; Engineering Economics; Applied Statistics.

TEACHING EXPERIENCE**Graduate Teaching Assistant and Guest Lecturer - Case Western Reserve University**

Jan 2023 - Present

Course development and guest lecturing

- ECSE 360/460 - Manufacturing and Automated Systems: Design and Optimization (Fall 2025): co-designed the MCDM portion of the curriculum and delivered lectures; supported group technology and clustering, facility layout, assembly-line balancing, location problems, reliability, and statistical quality control.
- ECSE 352 - Engineering Economics and Decision Analysis (Fall 2024 and Fall 2025): designed the risk-decision module and delivered multiple lectures on mean-variance analysis, decision trees, expected utility theory, and GDT; supported cash-flow modeling and economic-feasibility analysis.
- ECSE 350/450 - Operations and Systems Design (Spring 2026): supported forecasting, aggregate planning, inventory, scheduling, supply-chain and transportation modeling; led Excel Solver instruction, exam reviews, and project evaluation.

Additional teaching assignments

- ECSE 416 - Convex Optimization in Engineering (Spring 2024 and Spring 2025): supported convex formulation, solution methods, and MATLAB implementation.
- ECSE 346 - Engineering Optimization (Spring 2024 and Spring 2025): supported linear, network, dynamic, integer, and nonlinear programming.
- EPOM 405 / ECSE 411 - Applied Engineering Statistics (Spring 2026): supported probability, statistical inference, regression, ANOVA, and design of experiments.
- BUAI 445 - Advanced Marketing Analytics (Spring 2023): guided Python-based analytics and machine-learning projects for managerial decision making.
- ECSE 408 - Introduction to Linear Systems (Fall 2023): supported state-space analysis, stability, controllability, observability, feedback design, and MATLAB.

PROFESSIONAL EXPERIENCE**Product Development Manager - Olababy, Beaverton, OR**

Jun 2021 - Mar 2022

- Led development of an automatic baby-formula dispenser from market requirements through engineering validation: defined technical specifications, assessed feasibility, designed the PCB, and built an engineering-validation prototype.
- Coordinated design, engineering, marketing, and supplier-facing activities; translated customer and brand requirements into implementable specifications and supported supply-chain execution.

RESEARCH METHODS AND SOFTWARE

Optimization and decision science: Convex, linear, network, dynamic, integer, and nonlinear programming; MCDM and utility modeling; forecasting, inventory, scheduling, facility location, transportation, reliability, and quality control.

Methods and software: Out-of-sample validation, experimental design, sensitivity analysis, panel regression, Newey-West HAC, clustered standard errors, bootstrap inference, and reproducibility audits; Python (PyTorch, TensorFlow, scikit-learn, NumPy, pandas), MATLAB, C/C++, Excel Solver, LaTeX, and Git.